A3-PS027-TC439

IPv6 Website and Dependency Readiness Auditor

Problem Statement PS-027 • Area: Internet Measurements

Team SYNEXIS

Sajeena Anna Jacob | Aarcha Suresh | Dr. Rajesh K S (Faculty Member)

Saintgits College of Engineering

AIORI Hackathon 3 • October 13, 2026

Problem Description

A website that resolves over IPv6 is not necessarily a website that works over IPv6.

- **IPv6 is the long-term path for Internet growth.** IPv4 addresses are exhausted, and IPv6-only or IPv6-mostly networks (mobile, enterprise, cloud) are becoming common.
- **Readiness is usually checked at the front door only.** The main domain may have an AAAA record, while its nameservers, CDNs, fonts, analytics, APIs and script hosts are still IPv4-only.
- **One IPv4-only dependency can break a page** for an IPv6-only user (missing scripts, styles, login or payment). On dual-stack networks the failure stays hidden.
- **Existing checkers test one hostname at a time**, so site owners get no complete, actionable view of what blocks IPv6 for their site.
- **Our goal:** an automated auditor that scans a website **and its full dependency tree**, tests real IPv6 reachability, and reports a readiness score with a prioritised fix list.

Background

Readiness is a property of the whole dependency graph, not a single hostname.

- **IPv6** uses 128-bit addresses [1]. Most sites run **dual-stack**, publishing both A and AAAA DNS records [2].
- **Clients choose between addresses** using default address selection [3] and **Happy Eyeballs v2** [4], which races IPv6 against IPv4 and silently falls back, so broken IPv6 is easy to miss.
- **IPv6-only networks** (often with DNS64/NAT64 [5][6]) expose the gaps: IPv4-only hosts may still load through translation, but with extra latency and failure points.
- **A modern page loads dozens of third-party origins**, so one weak link can degrade the whole experience.
- **Internet measurement approach:** active probing (DNS, TCP, TLS, HTTP, page load) from an IPv6 vantage point gives evidence instead of assumptions.

IPv6 readiness chain: every link must hold

Literature & RFC Review

Standards define what to test; existing tools show the gap we fill.

Reference	Relevance to PS-027
RFC 8200 [1]	IPv6 base specification: defines correct IPv6 behaviour the auditor expects.
RFC 3596 [2]	AAAA records: the first thing the DNS audit checks, for the site and its nameservers.
RFC 6724 / 8305 [3][4]	Address selection and Happy Eyeballs: why failures are masked, so tests must force IPv6.
RFC 6146 / 6147 [5][6]	NAT64 / DNS64: models how IPv6-only clients reach IPv4-only dependencies.
RFC 8201 [7]	IPv6 Path MTU Discovery: large-response and blackhole failures to test for.
RFC 6540 [8]	BCP 177: IPv6 support expected on all IP-capable nodes; the policy basis for readiness.
RFC 9386 [11]	IPv6 deployment status: current adoption context and remaining barriers.
Measurement tools [9][10]	Google/APNIC publish adoption statistics; RIPE Atlas gives IPv6 probes; single-site checkers test one hostname. **Gap: no one-step tool audits a site plus all its dependencies.**

Proposed Solution

An automated six-stage pipeline that tests what an IPv6-only user would actually experience.

1	**Input**	A URL via web dashboard or API, or a CSV list of sites.
2	**DNS audit**	A/AAAA records for the site; nameservers have AAAA and answer over IPv6.
3	**IPv6 reachability**	Forced-IPv6 TCP (80/443), TLS handshake and HTTP fetch, compared with IPv4 (status, content hash, latency).
4	**Dependency discovery**	A headless browser loads the page and records every requested host: scripts, styles, fonts, CDNs, APIs, redirects.
5	**Dependency audit**	Each unique host gets the same DNS and reachability tests (parallel, cached).
6	**Score and report**	Readiness score 0-100, per-dependency table and fixes; export as PDF, HTML, JSON or CSV.

Stack: Python • FastAPI • Playwright • dnspython • httpx • asyncio • SQLite • web dashboard

Host status: NATIVE IPv6 | PARTIAL | IPv4-ONLY (needs NAT64) | UNREACHABLE / BROKEN

Optimization Proposed

Compared with typical manual, single-hostname checking:

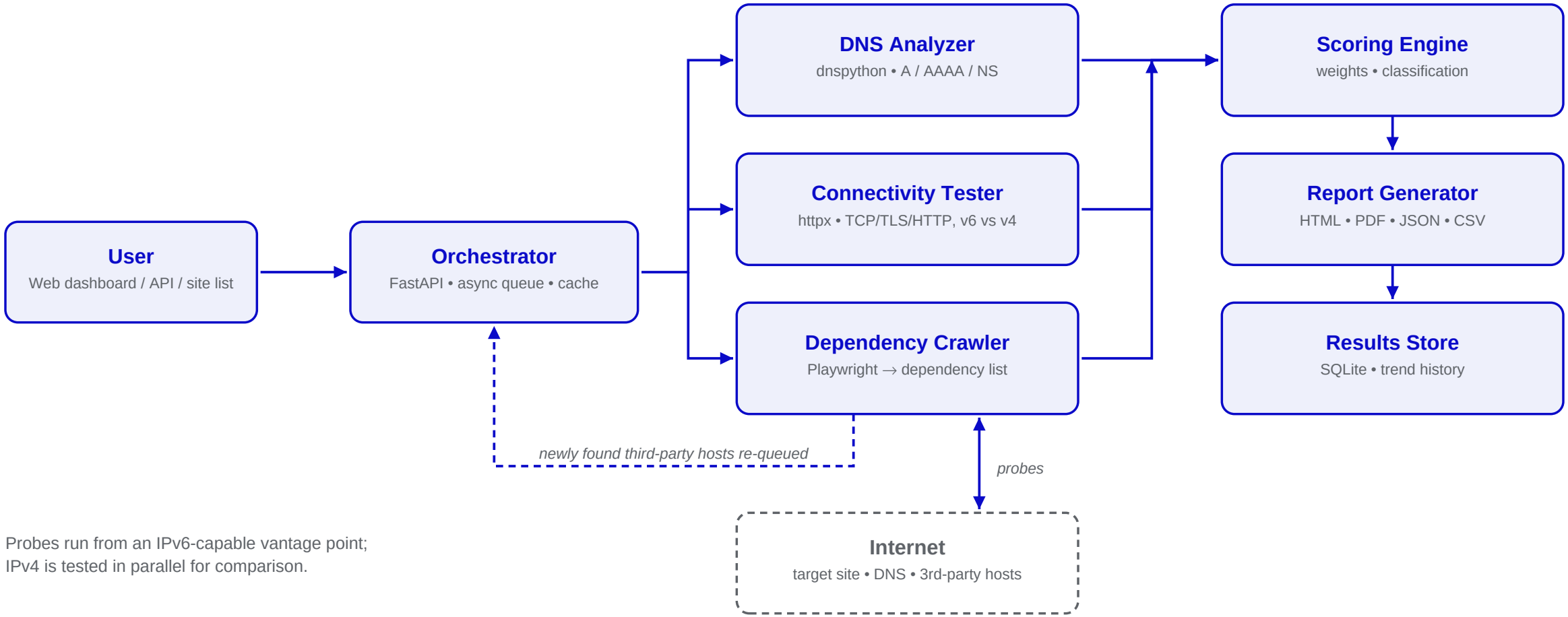
Aspect	Before (typical check)	After (our auditor)
Scope	Main hostname only	Full dependency tree of the page
“Ready” means	AAAA record exists	AAAA + working TCP/TLS/HTTP over IPv6 + content parity with IPv4
Test environment	Dual-stack hides failures	Forced-IPv6 tests and IPv6-only simulation
Speed	One site, by hand	Async parallel scans; shared CDNs checked once and cached
Accuracy	Single test, risk of false results	Retries; resources weighted by criticality (first-party, render-blocking)
Output	Raw pass/fail	Score 0-100, root cause and prioritised fixes
Scale and tracking	One-off	Batch mode for many sites; scheduled re-scans with history

Readiness score weights (total 100)

20	30	10	30	10
DNS	IPv6 reachability	Content parity	Dependencies	Robustness

Solution Architecture & Design

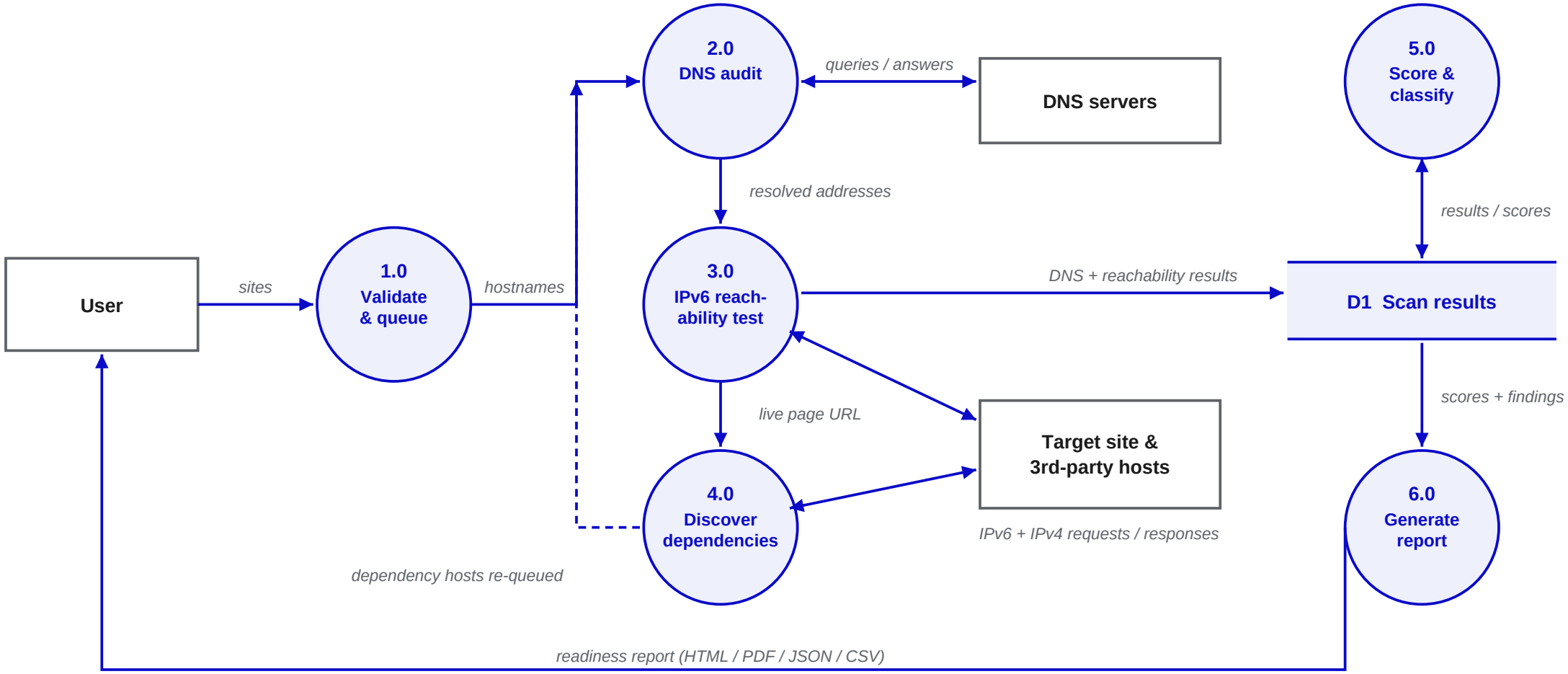
Component view of the auditor and how a scan flows through it.



Probes run from an IPv6-capable vantage point;
IPv4 is tested in parallel for comparison.

Data Flow Diagram

How data moves between the user, the scan processes and the internet.



Deployment Timeline

Agile one-to-two-week sprints, aligned with the hackathon milestones.

Phase	Dates (2026)	Deliverable
1. Setup and design	10 - 14 Oct	Repository, presentation and team intro video (Level 1 steps)
2. Core scanner	16 - 24 Oct	Crawler, DNS audit and forced-IPv6 connectivity test; mentoring sessions (Step 8)
3. Dependency engine	25 Oct - 4 Nov	Headless crawl, dependency audit and scoring; evaluator connect
4. Reports and testing	5 - 23 Nov	HTML/PDF reports, batch mode, validation on known IPv6-ready and IPv4-only sites
5. Regional demo	24 - 30 Nov	Live demo and results (if shortlisted)
6. Finals	17 Dec	Final polished tool and demo at IIT Jammu (if selected)

Methodology: build and test each stage on a seed set of sites with known IPv6 status, then scale to batch scans. Every sprint ends with a working demo.

Bibliography

[1] S. Deering and R. Hinden, “Internet Protocol, Version 6 (IPv6) Specification,” RFC 8200, IETF, Jul. 2017.
[2] S. Thomson, C. Huitema, V. Ksinant and M. Souissi, “DNS Extensions to Support IP Version 6,” RFC 3596, IETF, Oct. 2003.
[3] D. Thaler, R. Draves, A. Matsumoto and T. Chown, “Default Address Selection for Internet Protocol Version 6 (IPv6),” RFC 6724, IETF, Sep. 2012.
[4] D. Schinazi and T. Pauly, “Happy Eyeballs Version 2: Better Connectivity Using Concurrency,” RFC 8305, IETF, Dec. 2017.
[5] M. Bagnulo, P. Matthews and I. van Beijnum, “Stateful NAT64: Network Address and Protocol Translation from IPv6 Clients to IPv4 Servers,” RFC 6146, IETF, Apr. 2011.
[6] M. Bagnulo, A. Sullivan, P. Matthews and I. van Beijnum, “DNS64: DNS Extensions for Network Address Translation from IPv6 Clients to IPv4 Servers,” RFC 6147, IETF, Apr. 2011.
[7] J. McCann, S. Deering, J. Mogul and R. Hinden (Ed.), “Path MTU Discovery for IP version 6,” RFC 8201, IETF, Jul. 2017.
[8] W. George, C. Donley, C. Liljenstolpe and L. Howard, “IPv6 Support Required for All IP-Capable Nodes,” RFC 6540 (BCP 177), IETF, Apr. 2012.
[9] Google, “IPv6 Statistics,” and APNIC Labs, “IPv6 Capability Measurement.” Online measurement projects.
[10] RIPE NCC, “RIPE Atlas,” global Internet measurement platform, atlas.ripe.net.
[11] G. Fioccola et al., “IPv6 Deployment Status,” RFC 9386, IETF, Apr. 2023.

Thank You for Your Attention!

Team SYNEXIS • A3-PS027-TC439

IPv6 Website and Dependency Readiness Auditor • AIORI Hackathon 3

Saintgits College of Engineering